

Factors Associated with Depression Among U.S. Adults: Analysis of NHANES Data

Abstract

Background

Depression is a serious public health issue that leads to an inferior quality of life, increased risk of chronic diseases and a higher chance of disabilities and committing suicide. Determining factors related to depression is important for preventable and intervention approaches. This study examined demographic, social, economic and health determinants of depression among individuals residing in the United States.

Methods

This is a cross-sectional study design based on the data of National Health and Nutrition Examination Survey (NHANES) of 2021-2023 cycle. A sample of 6,337 adult individuals detailed information has been included. Depression was measured using Patient Health Questionnaire - 9 (PHQ-9). Various independent factors are included such as age, gender, race/ ethnicity, education, marital status, pregnancy status, income to poverty ratio, BMI, and survey month. Statistical analysis like summary statistics, bivariate analysis, multivariable logistic regression, kernel smoothing curves, Random Forrest, and Leave-one-covariate (LOCO) analysis were performed.

Results

The mean age of the individuals was 52.33 years and 55% of them were female participants. Logistic analysis indicated that higher BMI is associated with increased risk of odds of depression (OR=1.04, $p < 0.001$). whereas high income poverty ratio was correlated with lower risk of depression (OR=0.807, $p = 0.003$). Participants who were divorced had a higher risk of depression compared to the reference group (OR=1.68, $p = 0.018$). Summary statistics showed higher levels are among females, unmarried individuals and with low education levels. According to random Forrest and LOCO studies age, BMI, income, and education are all associated factors of depression.

Conclusion

Depression among individuals in United States is strongly associated to socioeconomic challenges, breakdown of marriage and obesity. Results emphasize the significance of combining mental health treatments with community support, prevention of chronic diseases, and specialized screening programs for high-risk groups.

Key words

Depression, NHANES, PHQ-9, BMI, obesity, income, marital status, mental health, education, logistic regression, Random Forrest, Loco Analysis

1.Introduction

Depression is one of the world's most common mental health problem, and it continues to be a serious public health concern. It can be identified by chronic sadness, less interest in daily life activities, fatigue, low concentration, insomnia, and less ability to perform daily tasks. Depression causes considerable damage, low standards of life, increased medical care usage, and premature death. According to the World Health Organization, around 280 million people are suffering from depression worldwide, accounting for about 3.8% of the total population (WHO,2023).

In the United States, over 21 million individuals suffered from at least one serious condition of depression in the preceding year (SAMHSA,2023). Depression is more common in women, young adults, low-income persons, and those with chronic medical conditions including diabetes, hypertension, and cardiovascular disease (Brody et al., 2018; Hasin et al.,2018). Regardless of the existence of effective treatment alternatives, many individuals go untreated due to discrimination, financial difficulties, and reduced access to mental health services (Mojtabai et al., 2011).

Earlier study evidence has repeatedly found that sociodemographic factors are significantly connected with risk of depression. Young people often report higher levels of depressive symptoms than older adults, which can be related to academic pressure, insecure work, financial difficulties, and social transformations (Kessler et al., 2003; Twenge et al., 2019). In contrast, depression in older individuals may be associated with being alone, grief, disabilities, and chronic problems (Blazer,2003).

Sex disparities are also generally reported, with women being twice as likely as males to experience depression in their entire lives (Albert,2015, Salk et al.,2017). Hormonal shifts, parenting obligation trauma exposure, and psychological and social stressors could all contribute to these differences. Marital status has also been proven to effect mental health, with married individuals experiencing lower levels of depression than widowed or separate individuals due to greater emotional and financial aid networks (Wade&Pevalin,2004).

Race and ethnic background are also connected with depression, though the conclusions are sometimes complex. Several research show that racial and ethnic minority communities may have higher levels of psychological distress because of structural disparities, racial prejudice neighbourhood deprivation, and limited access to quality medical care (Williams et al., 2007; Baiely et al., 2017). Cultural variations in describing symptoms and help seeking habits may also influence prevalence estimates. These findings emphasize the importance of investigating demographic inequalities in depression using nationally representative data.

Economic factors remain among the most powerful predictors of depression. Lorant et al. (2003) discovered that people with lower socioeconomic position were more likely to suffer from depression than those with greater incomes. Adults living below the poverty line often face food insecurity, low-quality housing, limited transportation, and persistent financial difficulties, all of which have a poor impact on mental health (Butterworth et al.,2012). In the United States, levels of depression among lower income individuals are about double as much as among higher income adults (Pratt&Brody,2014). Higher levels of education might act as a protective factor because it relates to better employment opportunities, more financial opportunity, improved educational knowledge, and improved access to health services

(Freeman et al., 2016). This suggests that financial status and educational attainment are important economic determinants of depression.

In addition to socioeconomic variables, body mass index (BMI) has gained popularity as related to health predictor of depression. BMI is a widely used measure of height and weight that classifies people as underweight, normal, and obese. Obesity affects more than 41.9% of US adults, making it a major public health problem (CDC,2023). Earlier research has found that obese adults are more vulnerable to experience depression than those of normal weight (Luppino et al.,2020). Obesity may cause depression due to chronic inflammation, metabolic dysfunction, sleep disturbance, body dissatisfaction, and societal judgement (Milaneschi et al., 2019). According to research, obesity raises the likelihood of future depression, while depression may increase body weight through emotional eating habits, sedentary behaviour, and pharmaceutical side effects (Faith et al., 2011; Mannan et al., 2016).

Large, nationally representative datasets offer significant possibilities to investigate these interactions across different communities. The National Health and Nutrition Examination Survey (NHANES) is continuous survey undertaken by the Centres for Disease Control and prevention to evaluate the health and nutritional status of the US population by using screenings, physical examinations, and standard laboratory tests. NHANES provides demographic information, BMI measures, and validated depression screening with the Patient Health Questionnaire-9 (PHQ-9), making it a trustworthy resource for population mental wellness investigation (Kroenke et al.,2001). Prior NHANES based studies found significant relationships among obesity and depression, poverty, cigarette smoking, physical inactivity, and chronic illness severity (Pratt & Brody,2014).

Although many prior studies have focused on specific indicators of depression few have looked at sociodemographic, economical, and BMI related factors together in one integrated framework using nationally representative data from the United States. Understanding how these factors interact to cause depression might aid in finding vulnerable groups such as low-income individuals, women, young adults, unmarried people, and obese adults. The goal of this study was to use NHANES data to investigate the relationship between sociodemographic variables, economic variables, economic variables, and body mass index with depression among people in the United States.

2.Methodology

2.1 Data

The study used data from the Centers for Disease Control and Prevention (CDC) 2021-2023 National Health and Nutrition Survey (NHANES). NHANES provides data analysis on healthcare access and illness prevalence, as well as nutritional status and other details about demographic for the non-institutionalized civilians people living in the United States. The data is cross-sectional, implying that each participant given data for all variables at once.

This analysis included three NHANES components: the Depression Screener Questionnaire (DPQ_L), the Demographic variables and Sample weights file (DEMO_L), and the Body measurements (BMX_L). These data sets have been merged using the participant's unique ID, SEQN. Participants with incomplete information on important characteristics and below the age of 20 were not taken into consideration. There were 6,337 adult participants in the final analytical sample.

2.2 Variables

The Patient Health Questionnaire -9 (PHQ-9) was used to measure the dependent variable, which was depression. Nine items form up the PHQ-9, which evaluates depressed symptoms during the preceding two weeks. Higher scores show more severe depressive symptoms. Total scores range from 0 to 27. Based on specified smallest requirements, a binary depression variable was developed.

Independent variables include population demographics, socio-economic status and health related factors that have been linked to depression. Demographic factors include Age, sex, and race or ethnic background. Socioeconomic factors consist of levels of education, marital status, and family income, as decided by the income-to-poverty ratio. Measurements of height and weight were used to construct body mass index (BMI), which was included as a health-related indicator.

BMI was chosen because prior research has shown correlations between depression and obesity via psychological, social, and biological factors. Inflammation, decreased physical activity, body dissatisfaction, and the burden of chronic diseases are all linked to higher BMI, and these factors may have an impact on mental health problems.

2.3. Analytical approach

The unique participant identification (SEQN) was used to combine the DEMO_L, DPQ_L and BMX_L datasets from NHANES. Participants with missing data on depression status or important variables were not included in the final analytical sample. Participants demographics were summarized using Descriptive statistics, with continuous variables shown as means with standard deviations and categorical variables provided as frequencies and percentages. Using chi-square tests for categorical data and suitable comparisons for continuous variables, bivariate analyses were carried out to investigate correlations among depression status and participant variables. Using Sociodemographic, economic and health related characteristics as independent variables and depression status as the dependent variable, multivariable logistic regression was used to find dependent predictors of depression. 95% Confidence Interval (Cis) and adjusted odds ratios (ORs) were shown. Additionally, age related depression patterns across specific subgroups were shown using kernel smoothing techniques, and the variable importance and predicting contribution of study variables were evaluated using Random Forest and leave-one-covariate out (LOCO) analyses. RStudio was used for all analyses, and $p < 0.05$ was considered statistically significant.

Table 1**Description of study variables used in the Analysis.**

Variable	NHANES code	Source File	Type of Variable	Analysis	Code/ Measurement
Depression status	PHQ-9	DPQ_L	Binary	Dependent Variable	YES = PHQ-9 > 10, No = PHQ-9 <10
PHQ-9 Total score	DPQ010-DPQ090	DPQ_L	Continuous	Outcome	Range: 0-27
Age Group	RIDAGEYR	DEMO_L	Continuous	Independent variable	20-29,30-39,40-49,50-59,60-69,>70
Sex	RIAGENDR	DEMO_L	Categorical	Independent Variable	Male, Female
Race	RIDRETH1	DEMO_L	Categorical	Independent Variable	Mexican American, Other Hispanic, Non-Hispanic white, Non-Hispanic Black, Other
Education level	DMDEDUC2	DEMO_L	Categorical	Independent Variable	< 9 th Grade, 9 th -11 th Grade, High school, associate degree, Bachelors
Marital status	DMDMARTZ	DEMO_L	Categorical	Independent Variable	Married, Widowed, Single
Income-to-Poverty Ratio	INDFMPIR	DEMO_L	Categorical	Independent variable	Higher values show, higher income status
Month of survey	RIDEXMON	DEMO_L	Categorical	Covariate	November-April, May-October
Pregnancy status	RIDEXPRG	DEMO_L	Categorical	Covariate	Pregnant, not pregnant, Unknown
Body Mass Index (BMI)	BMXBMI	BMX_L	Continuous	Independent	Kg/m ²

3. Results

The final analytical sample consists of 6,337 adults living in the United States. Individuals stood for a wide range of adult age groups, with a mean age of 52.33 ± 18.28 years. Of the sample, 55.0% were female, and 45.0% were male. While 52.9% of individuals were surveyed

between May and October, while 47.1% were surveyed between November and April. Non-Hispanic white includes the largest sample group by race/ ethnicity (58.0%), followed by Non-Hispanic Black (12.0%), another race is (12.0%), Other Hispanic (10.0%), and Mexican American (7.1%). As of education, 21% have completed their high school, while 35.0% had a bachelor's degree or above. 54% of the participants were married. The median body mass index (BMI) was $30 \pm 7 \text{ kg/m}^2$, the mean income to poverty ratio is 2.89 ± 1.66 , and the mean PHQ-9 depression score was 4.2 ± 5.0 .

(Table 2). Descriptive Statistics

Covariates	Count	Percentage	Depression Yes N%	Depression NO N%	P-value
Month of Survey					0.2
November - April	2987	47.1	49	46	
May- October	3350	52.9	51	54	
Gender					<0.001
Male	2858	45.0	37	47	
Female	3479	55.0			
Age					<0.001
20-29	1028	16.2	18.3	81.7	
30-39	1234	19.5	16.5	83.5	
40-49	1187	18.7	15.2	84.8	
50-59	1160	18.3	13.7	86.3	
60-69	964	15.2	11.6	88.4	
70 or above	764	12.1	10.1	89.9	
Ethnicity					0.035
Mexican American	448	7.1	8.4	6.6	
Other Hispanic	647	10	11	9.7	
Non- Hispanic white	3684	58	56	61	
Non-Hispanic Black	791	12	14	11	
Other	767	12	12	11	
Education level					<0.001
Less than 9 th grade	292	4.8	6.2	3.1	
9th-11thgrade	482	7.9	10	6.8	
Highschool	1297	21	25	20	
Associate degree	1846	30	36	30	
Bachelor's degree or above	2143	35	22	40	
Marital status					<0.001
Married	3277	54	38	56	
Widow	1536	25	31	24	
Single	1246	21	31	19	

Pregnancy status					0.05
Pregnant	41	3.6%	2.2	4.1	
Not pregnant	1065	94	94	94	
Unknown Pregnancy status	28	2.5	3.9	1.4	
Age (Years)	52.33 ± 18.28				
Income-poverty ratio	2.89 ± 1.66				
Body Mass Index (BMI)	30 ± 7				
Depression score (PHQ-9)	4.2 ± 5.0				

Bivariate Analysis

Several statistically significant variations in depression state across participant variables were found by bivariate analysis. Depression was significantly more common in women than in men 63% of participants had depression, compared to 37% in men ($p < 0.001$). There were also notable age-related disparities, with younger individuals having a higher prevalence of depression than older adults. Individuals 70 years of age or older has the lowest prevalence of depression ($p < 0.001$), while participants aged 20-29 has the highest proportion of depression (Table 2).

Depression was correlated with race/ethnicity ($p = 0.035$). Compared with several other racial/ethnic groups, Mexican American and Non-Hispanic Black individuals had comparatively higher rates of depression. Depression was significantly correlated with educational level ($p < 0.001$). Individuals with a bachelor's degree or greater has the lowest levels of depression, while individuals with low education had higher rates. Additionally, there was a strong correlation ($p < 0.001$) between marital status and depression. Participants who were married showed a lower prevalence of depression than those who were widowed or single. The pregnancy status was statistically marginally significant ($p = 0.050$). Despite this, there was no significant correlation between the month of survey participation and depression ($p = 0.200$).

Multivariable logistic Regression

Multiple factors continued to be independently linked to depression even after controlling for socioeconomic, demographic, and healthcare related variables. There was a significant correlation between depression and marital status. Divorced people had 1.68 times higher chances of depression than the reference group (OR = 1.68, 95% CI: 1.09-2.59, $p = 0.018$). Additionally, individuals who are widowed had higher levels of depression (OR = 1.64, 95% CI: 0.93-2.83), however this correlation was not statistically significant ($p = 0.083$).

The income-to-poverty ratio showed a protective effect that was statistically significant. The risks of depression decreased by almost 19.3% for every unit rise in the income-to-poverty ratio (OR=0.807, 95% CI: 0.698-0.934, $p = 0.003$). This result suggests that people with more

financial resources had lower rates of depression. Depression and BMI were also strongly linked. The odds of depression increased by 4% for every unit rise in BMI (OR = 1.04, 95% CI: 1.02-1.06, $p < 0.001$), showing an independent relationship between higher body weight and a higher risk of depression.

In the fully adjusted model, depression was not statistically significant associated with age, race/ethnicity, education level, month of survey participation, or pregnancy status ($p > 0.05$). Wide confidence intervals suggested inadequate accuracy even if certain groups had higher odds ratio. Overall, the final regression model found that the best independent determinants of depression in American adults were BMI, marital status, and socioeconomic status.

Table 3: Multivariate Logistic Regression

Variable	Odds ratio	95% CI Lower	95% CI Upper	P-value
Intercept	0.053	0.004	0.47	0.014
Month of survey				
May- Oct	1.32	0.904	1.95	0.151
Race/ Ethnicity				
Non-Hispanic Black	0.528	0.238	1.18	0.116
Mexican American	1.08	0.568	2.15	0.817
Non-Hispanic Asian	0.673	0.293	1.54	0.347
Other	0.749	0.329	1.71	0.49
Education Level				
9th-11thGrade	2.38	0.477	19.1	0.339
Highschool	2.07	0.441	16.0	0.41
Associate degree	1.98	0.434	15.1	0.431
Bachelor's or above	1.31	0.274	10.2	0.763
Marital status				
Widowed	1.64	0.929	2.83	0.083
Divorced	1.68	1.09	2.59	0.018
Pregnancy status				
Not pregnant	1.78	0.633	6.44	0.32
Unknown pregnancy status	4.45	0.957	23.0	0.061
Age (years)	0.983	0.954	1.01	0.238
Income – poverty ratio	0.807	0.698	0.934	0.003
Body Mass Index (BMI)	1.04	1.02	1.06	< 0.001

Kernel smoothing Analysis.

Figure 1: SEX VS DEPRESSION

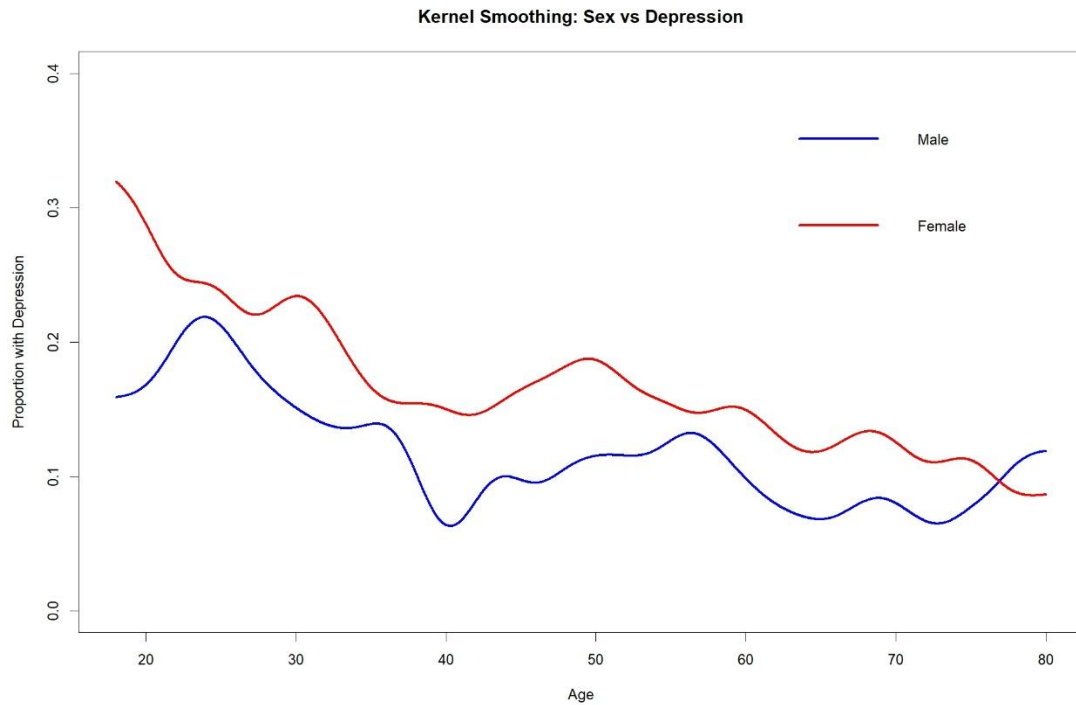


Figure 2: OBESITY VS DEPRESSION

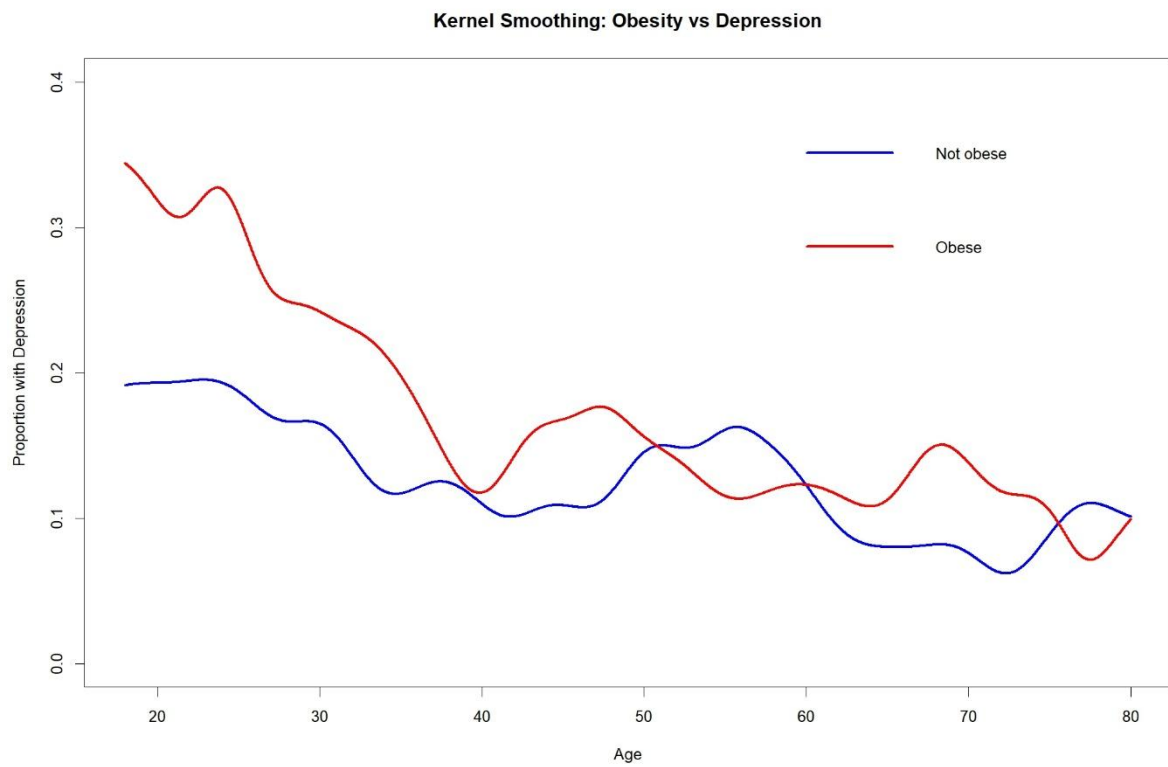


Figure 3: ICOME GROUP VS DEPRESSION

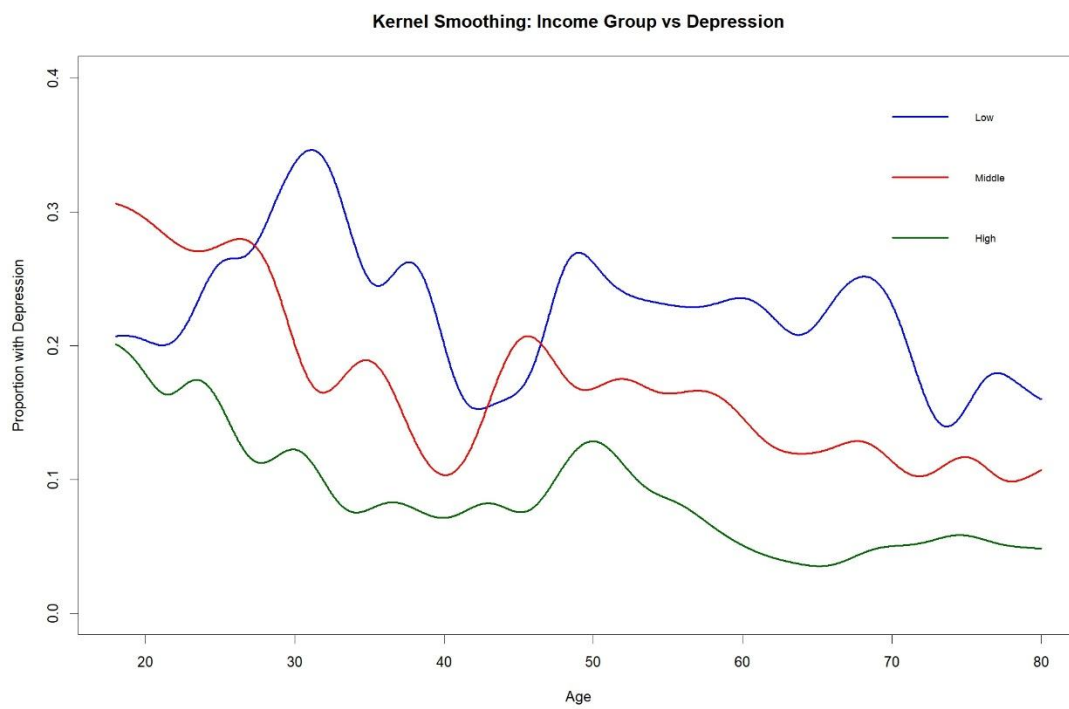


Figure 4: MARITAL STATUS VS DEPRESSION

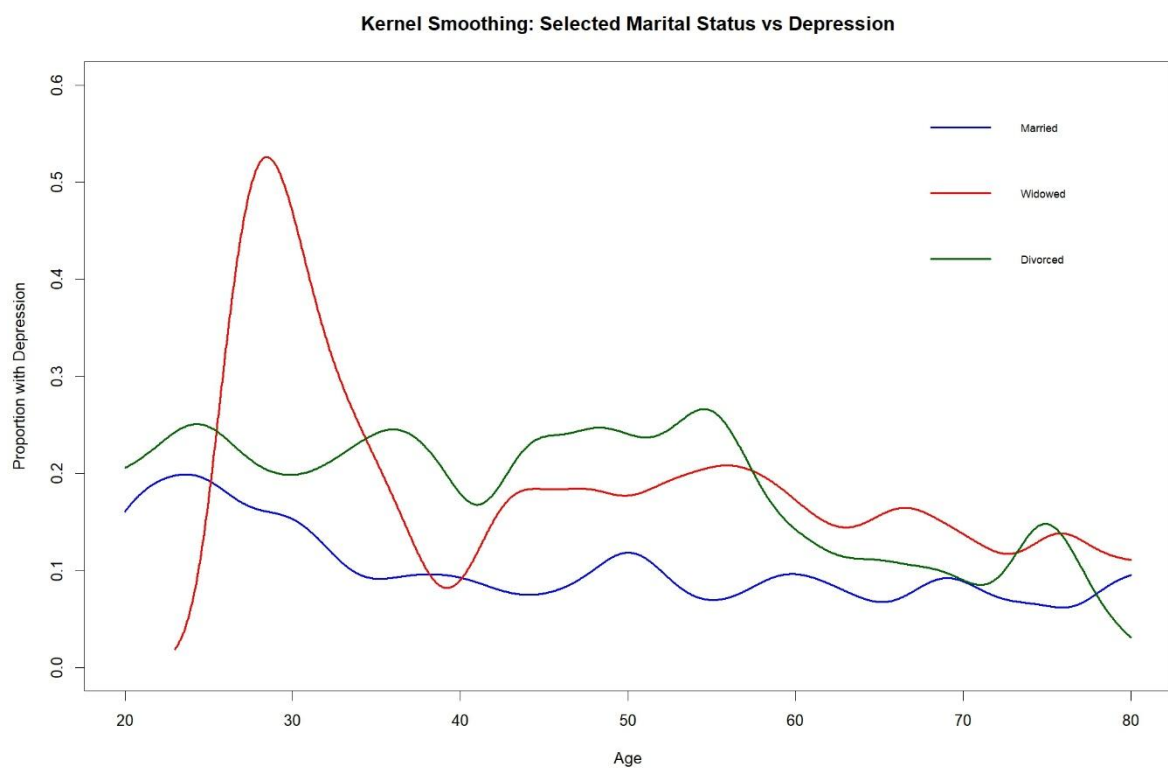
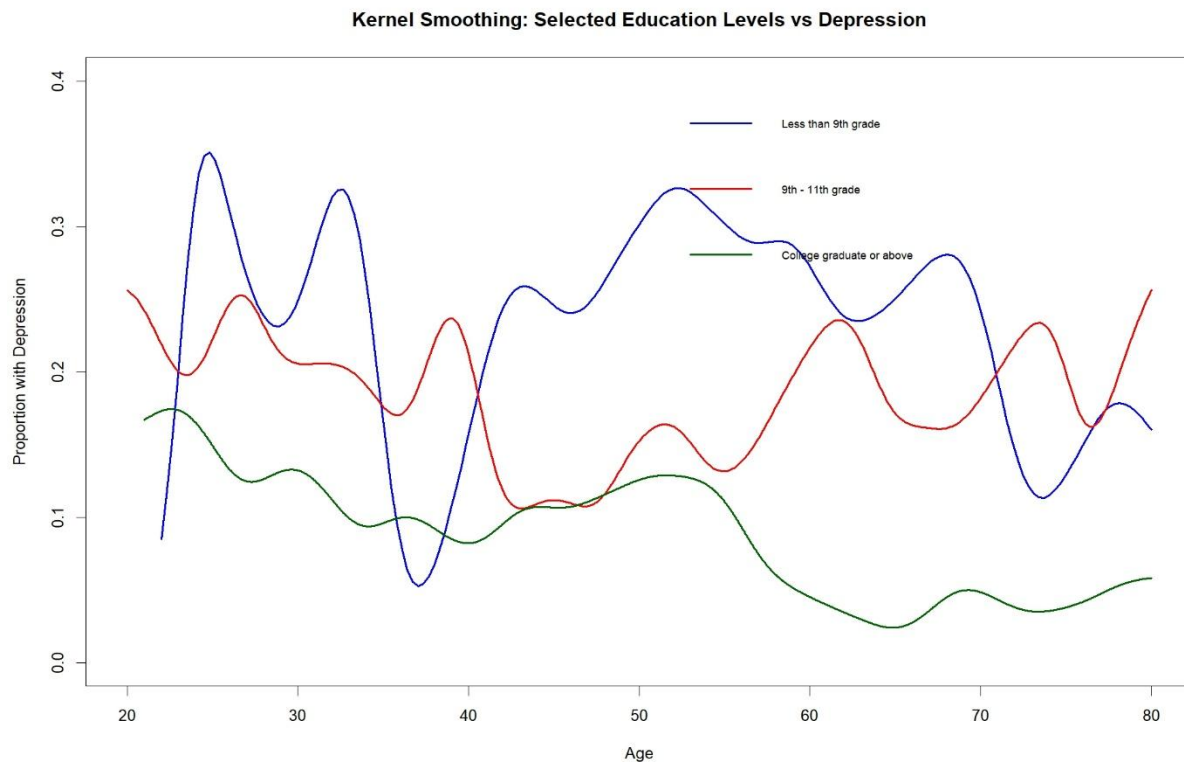


Figure 5: EDUCATION VS DEPRESSION



Age related depression patterns in some selected subgroups were assessed using kernel smoothing plots. These plots showed that depression was more common in women than in men in most age groups, especially in younger and middle adulthood, and that prevalence decreased with age. Additionally, obese people particularly younger adults were shown to have higher levels of depression in older age groups the differences decreased.

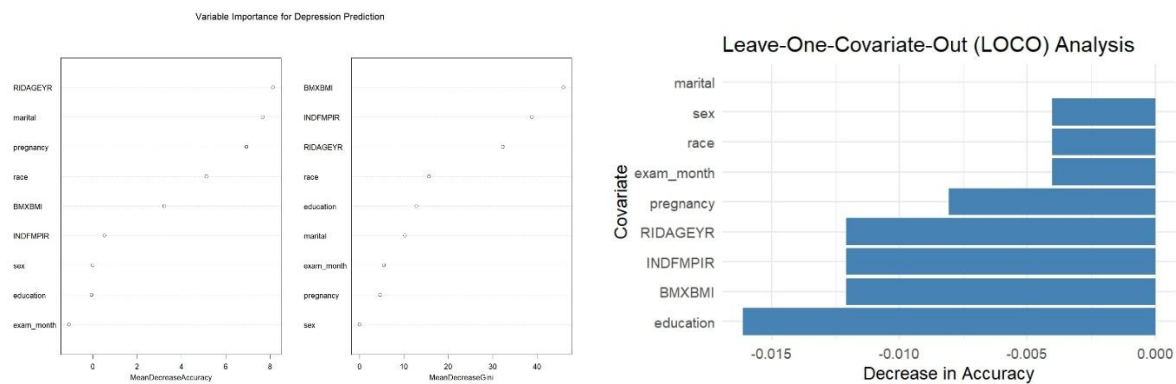
High income individuals had the lowest prevalence of depression, while lower income individuals continuously had greater levels, according to income-based smoothing curves. Across all age groups, married people had reduced rates of depression, while separated, widowed, and divorced people had greater rates. Participants with low education levels have higher rates of depression, while those with degrees had reduce rates, according to educational patterns. There are few seasonal trends because the curves for November- April and May-October were the same for all age groups.

Overall, the results of the smoothing analyses confirmed the significance of obesity, marital stability, and socioeconomic status in depression risk.

Random Forest Variable Importance and LOCO Analysis

To further assess the relative importance of variables to depression categorization, Random Forest Variable and LOCO Analysis techniques were conducted. Age and marital status were

the most crucial factors in the Random Forest model, according to Mean Decrease Accuracy, showing that eliminating these predictors would significantly lower model performance. While sex, education level, and survey month significantly less to overall prediction accuracy, status of pregnancy and race/ ethnicity shows significant predictive value.



The Mean Decrease Gini results, on the other hand, showed that the most crucial factors for enhancing node separation in the Random Forest model were body mass index (BMI) and income to poverty ratio. Additionally, age continued to be extremely significant under this measure, showing its continuous contribution across several approaches to variable importance assessment. In terms of lowering categorization impurity, gender and pregnant status seemed to be less significant. Overall, these results imply that the Random Forrest model's most significant predictors of depression were age, BMI, income, and marital status.

By analysing the decrease in model accuracy following in the removal of one predictor at a time, the LOCO Analysis offered further insight into the role of each individual covariate. This research revealed that education was a significant element in the model and that its omission resulted in the largest drop in predicted accuracy. In an analogous way, eliminating, BMI, age, income to poverty was associated to a significant decline in accuracy, showing their high predictive significance. On the other hand, the model's accuracy barely changed when survey month, gender, ethnicity, and marital status were removed, showing a smaller influence on the model's overall performance.

By combined, the Random Forest and LOCO models show that the most significant predictors of depression were age, education, BMI, and income, pregnancy status and marital status had moderate significance, and gender, race, and survey month had comparatively lower predictive value. These results confirm the more general conclusion that health related and socioeconomic factors strongly influence the likelihood of depression in adults living in United States.

4. Discussion

This study used data from NHANES to investigate the relationship between depression and sociodemographic economical, and health related characteristics among adults living in United States. After adjusting for multiple variables, the results proved that the body mass index (BMI), income to poverty ratio, and marital status were the most effective independent predictors of depression. These findings imply that social and physical health concerns have

an impact on depression in addition to psychological aspects (Katon, 2011; Pratt & Brody 2014).

The correlation among depression and marital status was one of the main outcomes. Depression was far more common in divorced people, and it was also more common in widowed individuals. These results are in supporting with earlier research showing that lower mental health outcomes are linked to marital disturbances, widowed, loneliness and decreases emotional support (Wade & Pevalin, 2004; Williams et al., 2007). Marriage may offer companionship, financial security, and emotional support all of which may protect against depression.

Depression was also significantly correlated with economic conditions. Higher income to poverty ratios were associated with decreased possibilities of depression, suggesting that having more income may be a protective sign of mental health. These results align other research that showed a connection between depressive symptoms and financial stress (Lorant et al., 2003; Butterworth et al., 2012). Psychological suffering may increase because of limited access to mental health treatments, stable housing, nutritious food, and medical services. These Findings highlights the complex interaction between socioeconomic disadvantage, physical health, and mental health, emphasizing the need for integrated public health interventions.

Depression and BMI were strongly associated the likelihood of depression, an increase of one unit in BMI. This outcome is following earlier research showing an equal link between depression and obesity (Luppino et al., 2010; Milanese et al., 2019). While depression may lead to weight gain through depressive eating patterns and lack of physical activity, obesity may increase the risk of depression through chronic inflammation, inadequate sleep, decreased mobility, physical dissatisfaction, and discrimination (Faith et al., 2011).

Descriptive and LOCO analysis revealed that age, gender, race, levels of education all are associated to the risk of depression even though they were not statistically significant in the adjusted regression model. Age and education were found to be significant predictors in Random Forest and LOCO analyses, and women experience an elevated level of depression than men. These results aligned with research proving socioeconomic disparities across age groups and a greater frequency of depression among women (Albert, 2015; Salk et al., 2017). Targeted screening and early intervention among high-risk groups such as low income and unmarried individuals can significantly reduce the burden of depression.

5. Strength and Limitations

This study has several strengths and limitations. Strengths include the use of NHANES sample, consistent data collection techniques, valid PHQ-9 standardized depression screening, objective determined BMI, and the application of several analytical techniques such as logistic regression and predictive modelling (Kroenke et al., 2001). Although some parameters were reported by individuals, depression was determined based on a screening tool rather than a clinical diagnosis, the cross-sectional design limits the possibility of causal inference, and unknown confounding factors like regular physical activity, sleep quality, history of trauma, and social support could still exist (Mojtabai et al., 2011).

6. Findings

The results have significant implications for public health. Lower income individuals. Adults experiencing marital disturbance and individuals experiencing higher BMI should be given priority in depression prevention initiatives. The burden of depression may be altered by increasing access to cost effective and affordable counselling services, fostering community support networks, and incorporating mental health screenings into primary care (World health Organization,2023).

7. Implications of Future research

Future research understands the casual relationship among obesity, poverty, marital disturbance, and depression further research should make use of longitudinal data. Further investigations should aim at mediating factors such a lifestyle choices, stress, inflammation and sleep quality and access to healthcare. Predictive models in the future might potentially ease finding elevated risk populations for early intervention.

8. Conclusion

The study concluded that the most significant indicators of depression among American adults were lower income, greater BMI, and marital instability. These results highlight how social, economic, and physical health issues influence depression. Public health initiatives that reduce financial difficulties, enhance healthy body weight management, and enhance social support networks may be crucial in lowering the prevalence of depression in the United States.

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